

Lab 6

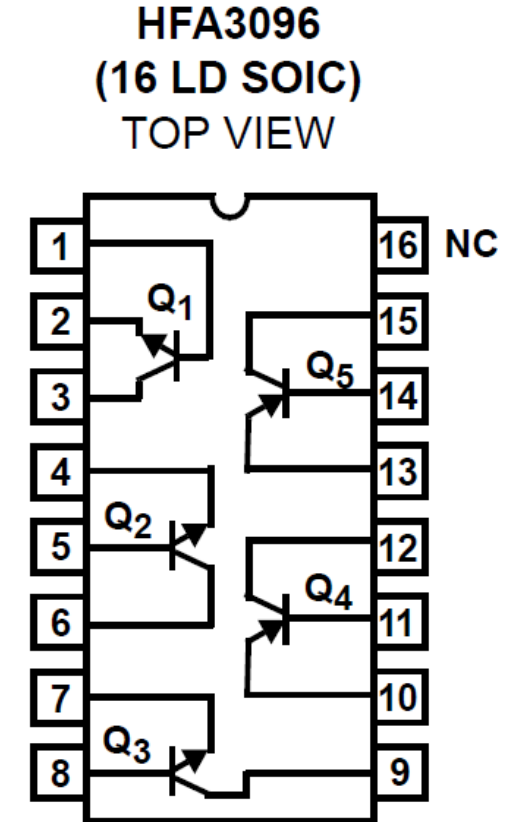
topics

5L.1,5L.1.1,5L.1.4, 5L.1.5

5L.1.1

In this part, we just setup the circuit but do not yet measure anything

- Google the datasheet for CA3096 to get the pinout. Be very careful where you put the ic on your breadboard (why?)
- Get checked off after 5L.1.1. At this point, **you have not yet connected** IN_A and IN_B (hint: they will NOT be connected directly to the function generator)
- Use the power buses, one pair for +15 and ground, the other for -15 and ground. This makes it easier to put a cap between each supply and ground to help prevent oscillation (and this circuit does tend to oscillate). You will also need to access the power supply ground when you are setting up the scope probes.
- in figure 5L.1.1, the circled numbers are the pin numbers on the IC



Keep going... more on next page...

Setup the input circuit and Check the transformer

In this section **DO NOT ATTACH the transformer TO THE CIRCUIT YET!!!! We are checking the transformer before attaching it.**

- Please interpret the one sentence in 5L.1.2 to mean “there are two ways to setup the inputs, described in 5L.1.3 and 5L.1.4. In this class, we will use 5L.1.4”
- **DO NOT DO 5L.1.3, instead do 5L.1.4. (see figure 5L.7)** The function generator attached to the primary makes the “difference signal”. The one attached to the center tap the common mode. Note that this lab uses both the function generator internal to the scope and the external function generator. Since we only have 2 channels on the scope, you’ll need to change what are connected to them for different measurements.
- Use the external function generator for the differential input. Its ground should only connect to the transformer primary and not to the other grounds to prevent ground loops
- **Check carefully that the output of the transformer is behaving as expected. Especially make sure the two outputs are 180 degrees out of phase for a large differential signal when the middle pin (common mode input) is shorted to ground instead of being attached to the common-mode function generator. Measure the attenuation of the differential signal from one side of the transformer to the other. (try removing the common mode input and shorting the middle pin to the differential signal function generator ground to see it better).**
- Start with both function generators with a 1kh sine of about the same amplitude
- Look at the 2 outputs **A and B** from the transformer. Show that you can adjust their common mode and differential signal separately. Show when you do this the output behaves as expected.

5L.1.5

Now attach the input to your circuit

- Start with both function generators with a 1kh sine of about the same amplitude
- This circuit tends to oscillate. Works best when probe on the input is on pin 1, not pin 5. note pin5 is the side with the gain.
- Measure the common mode gain and compare to the predicted value (see the lecture). **Note Fig 5L.8 is misleading. For common mode, the two inputs are in phase, the output is 180 degrees out of phase.**
- For the differential gain, on page 223, where it says “cutting common-mode amplitude to a minimum”, instead remove the connection to the center tap and connect it to ground.
- For the differential gain, The gain is awfully large (factor 40). Note that the transformer decreases the amplitude by about a factor of 20, so take this into account. For me, on the input to the transformer, a voltage of 2.6 V worked well.
- After reassembling to have both the common mode and differential, watch what happens to the signal as you dial the “common mode” signal from its minimum up to several volts. What do you see?
- When asked to use the current source, do the JFET version. The part number is E-152 (instead of the one in the text). Since this is a 1.5mA diode instead of the 0.75mA diode discussed in the text, just use one of them. See the current-regulating-diode.pdf in the datasheets in elms. For this part, for the common mode, try making the input amplitude as large as 10V
- Try to explicitly measure the common mode rejection. Try swapping the resistor and the diode and looking at the result visually as you increase/decrease the common mode input. Write some text on what you see
- **To avoid bending an ic's pins when removing it, pull it straight up (pull on BOTH sides at once)**

